

LAB EXPERIMENT RECORD

SET 15 — Product Sales Analysis

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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A. Aim

To analyze monthly product sales using R by creating a grouped bar chart, stacked area chart, and sales data table, and to develop a Tableau dashboard combining all three visualizations.

B. Dataset

Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400

2	Product B	1200	1400	1500
3	Product C	1100	1800	1600

Program 1 — Grouped Bar Chart

Aim

To create a grouped bar chart showing the sales of each product for the first quarter.

R Code

```
# Monthly Product Sales
```

```
sales <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1200, 1100),
  February = c(2200, 1400, 1800),
  March = c(2400, 1500, 1600)
)
```

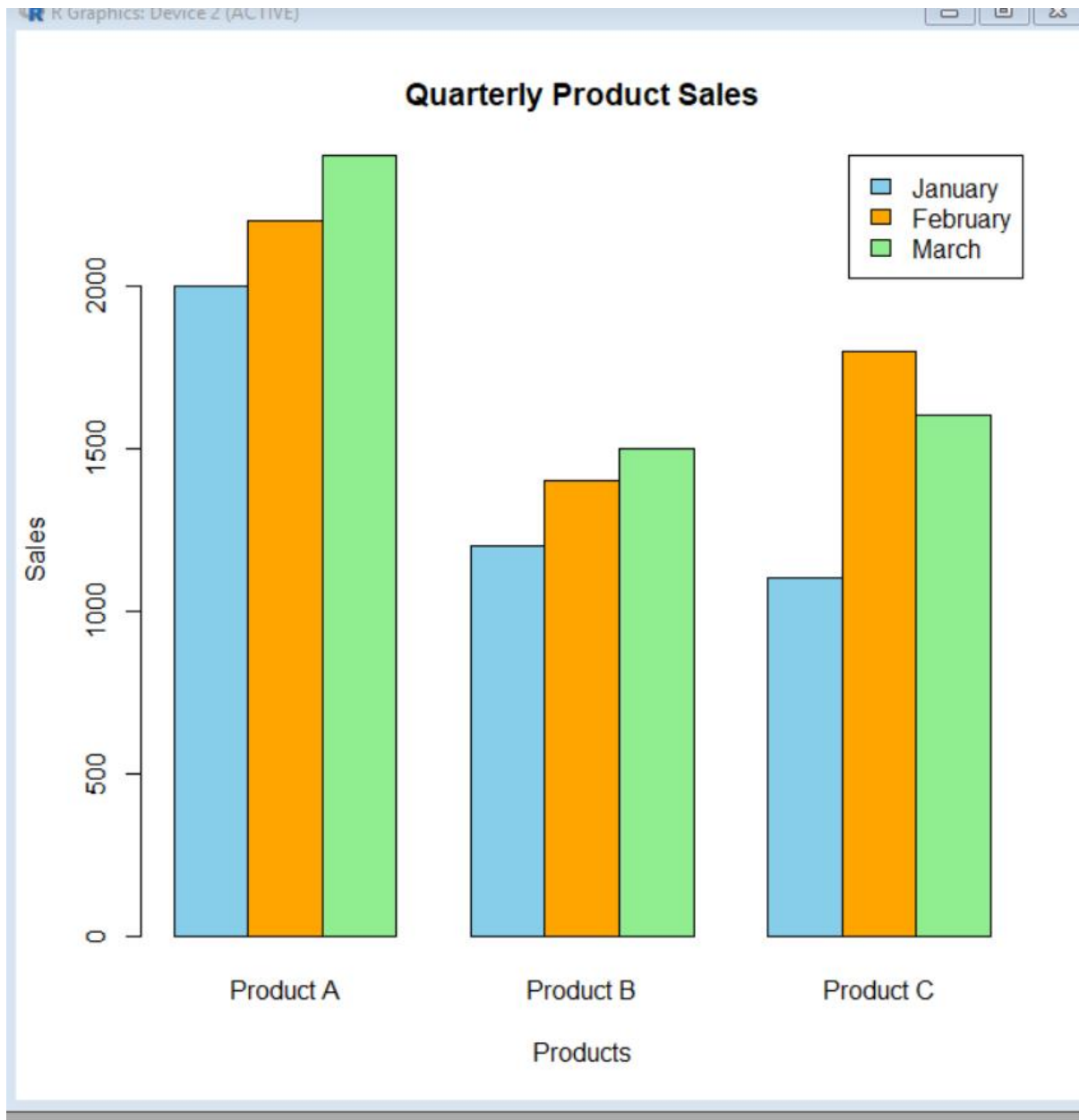
```
# Convert sales into a matrix
```

```
sales_matrix <- as.matrix(
  sales[, c("January", "February", "March")]
)
```

```
rownames(sales_matrix) <- sales$Product
```

```
# Create grouped bar chart
```

```
barplot(  
  t(sales_matrix),  
  beside = TRUE,  
  main = "Quarterly Product Sales",  
  xlab = "Products",  
  ylab = "Sales",  
  col = c("skyblue", "orange", "lightgreen"),  
  legend.text = c("January", "February", "March"),  
  args.legend = list(x = "topright")  
)
```



Observation

The grouped bar chart compares the monthly sales of all three products. Product A has the highest sales in each month, while Product B has the lowest sales in each month.

Program 2 — Stacked Area Chart

Aim

To create a stacked area chart representing the overall sales trend of all products over three months.

R Code

```
# Install package if required
# install.packages("ggplot2")

library(ggplot2)

# Monthly Product Sales

sales <- data.frame(
  Month = c("January", "February", "March"),
  Product_A = c(2000, 2200, 2400),
  Product_B = c(1200, 1400, 1500),
  Product_C = c(1100, 1800, 1600)
)

# Convert data to long format

sales_long <- data.frame(
  Month = rep(sales$Month, 3),
  Product = rep(
    c("Product A", "Product B", "Product C"),
    each = 3
  ),
)
```

```

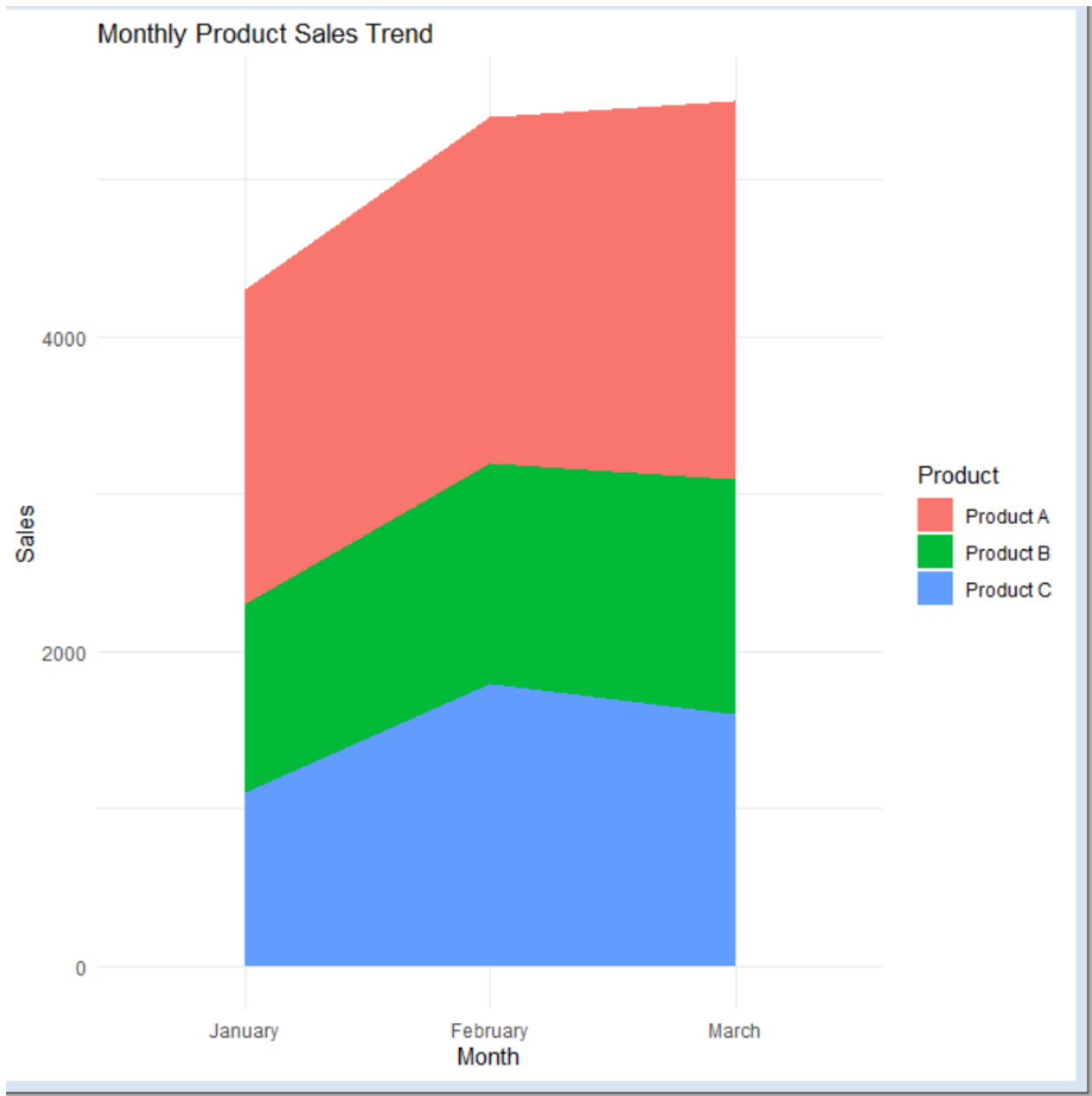
Sales = c(
  sales$Product_A,
  sales$Product_B,
  sales$Product_C
)

sales_long$Month <- factor(
  sales_long$Month,
  levels = c("January", "February", "March")
)

# Create stacked area chart

ggplot(
  sales_long,
  aes(
    x = Month,
    y = Sales,
    fill = Product,
    group = Product
  )
) +
  geom_area() +
  labs(
    title = "Monthly Product Sales Trend",
    x = "Month",
    y = "Sales",
    fill = "Product"
  ) +
  theme_minimal()

```



Observation

The stacked area chart shows the combined sales trend across the three months. Total sales increased from 4300 in January to 5400 in February and 5500 in March.

Program 3 — Monthly Sales Data Table

Aim

To create a table displaying monthly sales data for each product.

R Code

```
# Monthly Product Sales Data
```

```
sales <- data.frame(  
  Product_ID = c(1, 2, 3),  
  Product_Name = c(  
    "Product A",  
    "Product B",  
    "Product C"  
  ),  
  January_Sales = c(2000, 1200, 1100),  
  February_Sales = c(2200, 1400, 1800),  
  March_Sales = c(2400, 1500, 1600)  
)
```

```
# Display sales table
```

```
print(sales)
```

```
print(sales)  
Product_ID Product_Name January_Sales February_Sales March_Sales  
      1      Product A         2000          2200          2400  
      2      Product B         1200          1400          1500  
      3      Product C         1100          1800          1600
```

```
.
```


Observation

The table displays the monthly sales of Product A, Product B, and Product C for January, February, and March.

Program 4 — Tableau Interactive Dashboard

Aim

To develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and sales table for interactive exploration of product sales.

Step 1 — Prepare the Excel Dataset

Enter the following data into Excel and save it as Monthly_Product_Sales.xlsx.

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1200	1400	1500
3	Product C	1100	1800	1600

Open Tableau and connect to this Excel file.

Step 2 — Create Grouped Bar Chart

1. Open a new worksheet.
2. Rename it Grouped Bar Chart.

3. Drag Product Name to Columns.
4. Drag Measure Values to Rows.
5. Drag Measure Names to Color.
6. In the Measure Values card, keep only January Sales, February Sales, and March Sales.
7. Drag Measure Names to Filters and select only those three sales measures.
8. Select Bar in the Marks card.
9. Ensure the bars are displayed side by side rather than stacked.
10. Turn on labels if required.

Step 3 — Create Stacked Area Chart

1. Open a new worksheet.
2. Rename it Stacked Area Chart.
3. Drag Measure Names to Filters.
4. Keep only January Sales, February Sales, and March Sales.
5. Pivot the three monthly sales columns in the Data Source tab, if needed, to create Month and Sales fields.
6. Drag Month to Columns.
7. Drag Sales to Rows.
8. Drag Product Name to Color.
9. Change the Marks type to Area.
10. Ensure stacking is enabled.

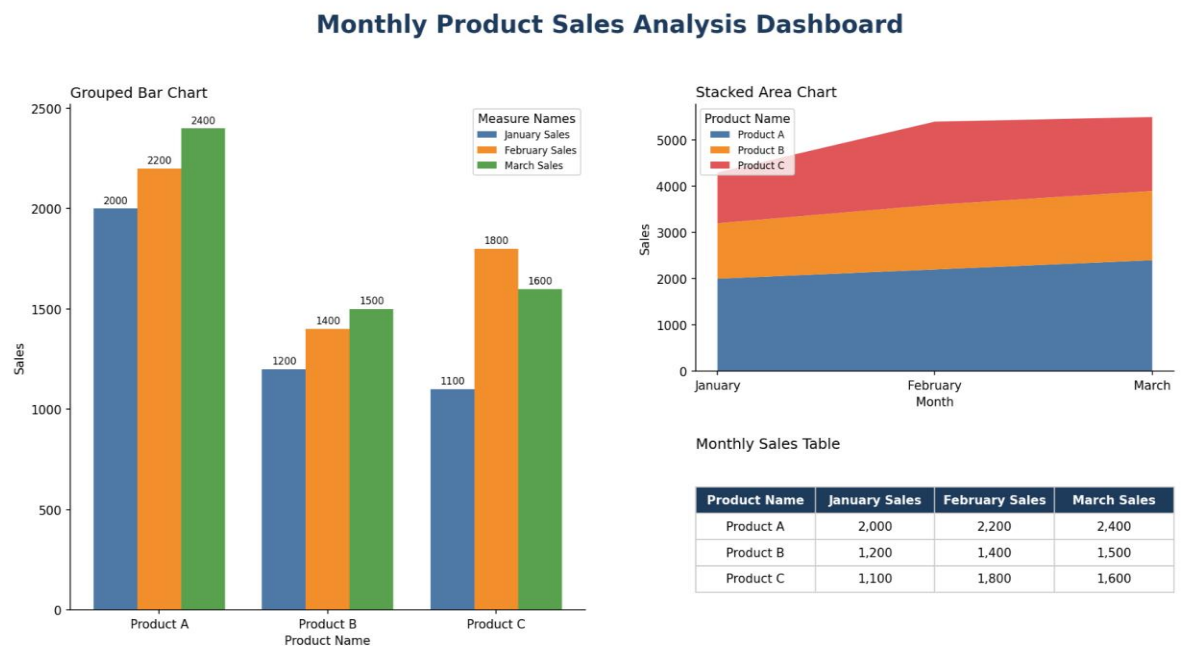
Step 4 — Create Sales Data Table

1. Open a new worksheet.
2. Rename it Monthly Sales Table.
3. Drag Product Name to Rows.
4. Drag Measure Names to Columns.
5. Drag Measure Values to Text.
6. Keep only January Sales, February Sales, and March Sales.

7. Check that all monthly sales values are displayed.

Step 5 — Create Tableau Dashboard

1. Click Dashboard → New Dashboard.
2. Drag Grouped Bar Chart onto the dashboard.
3. Drag Stacked Area Chart onto the dashboard.
4. Drag Monthly Sales Table onto the dashboard.
5. Arrange the charts and table neatly.
6. Add the title Monthly Product Sales Analysis Dashboard.
7. Add filters if required.
8. Save the Tableau workbook.



Observation

The dashboard combines the grouped bar chart, stacked area chart, and monthly sales table. It allows users to compare product sales across the first quarter.

C. Overall Procedure

1. Create the monthly product sales dataset.
2. Store the January, February, and March sales values.
3. Create a grouped bar chart using R.
4. Create a stacked area chart to visualize monthly sales trends.
5. Display the sales data in a table.
6. Import the dataset into Tableau.
7. Create the grouped bar chart worksheet.
8. Create the stacked area chart worksheet.
9. Create the monthly sales table worksheet.
10. Combine all three worksheets into a dashboard.
11. Add the dashboard title and labels.
12. Save and analyze the completed dashboard.

D. Result

The monthly product sales data was successfully analyzed using R and Tableau. The grouped bar chart, stacked area chart, and sales data table were created and combined into an interactive Tableau dashboard.

The analysis shows that Product A has the highest sales in all three months. Total sales increased from 4300 in January to 5500 in March, indicating an overall increase in sales during the first quarter.

E. Key Functions Used

Function / Feature	Purpose
data.frame()	Creates the sales dataset
as.matrix()	Converts sales data into a matrix
barplot()	Creates the grouped bar chart
library(ggplot2)	Loads the visualization package
geom_area()	Creates the stacked area chart
labs()	Adds chart titles and labels
theme_minimal()	Applies a clean chart theme
print()	Displays the sales table
Measure Names	Identifies monthly sales measures in Tableau
Measure Values	Displays sales values in Tableau

Marks → Color	Distinguishes products
Dashboard → New Dashboard	Creates the interactive dashboard